COURSE PROJECT REPORT

AI-powered Drone Delivery Management Platform

Course Code: 012012210504

Group members:
1. Võ Trần Gia Phụng
2. Bùi Khánh Chi
3. Trần Nguyễn Quỳnh Anh
4. Lê Chí Bảo
5. Mai Hoàng Long

Supervisor: Ths. Nguyễn Văn Chiến

Ho Chi Minh, August 2026

Acknowledgement

We would like to express our sincere gratitude to our lecturer for providing valuable guidance and support throughout this course.

We are also thankful to all group members for their dedication, effort, and continuous collaboration, which made the completion of this project possible.

Thank you very much.

Definition And Acronyms

Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
BPMN	Business Process Model and Notation
ERD	Entity Relationship Diagram
ETA	Estimated Time of Arrival
REST	Representational State Transfer
SRS	Software Requirements Specification
UI	User Interface
UX	User Experience
ERD	Entity Relationship Diagram

List of Tables

Contents

1 Project Overview

1.1 Project Information

- **Project Name:** AI-powered Drone Delivery Management Platform
- **Course Name:** Software Engineering
- **Course Code:** 012012210504
- **Members Information:**

Members	Role	Email
Nguyễn Văn Chiến	Supervisor	chiennv@ut.edu.vn
Võ Trần Gia Phụng	Leader	phungvtg4265@ut.edu.vn
Bùi Khánh Chi	Members	chibk0480@ut.edu.vn
Trần Nguyễn Quỳnh Anh	Members	anhtnq2727@ut.edu.vn
Lê Chí Bảo	Members	baolc897742@ut.edu.vn
Mai Hoàng Long	Members	longmh422679@ut.edu.vn

1.2 Current Situation

With the rapid growth of the low-altitude economy, drone delivery has emerged as a promising logistics solution due to its speed, flexibility, and ability to reach remote areas. The global drone delivery market is projected to reach USD 33.4 billion by 2030, with a 37.2% compound annual growth rate (CAGR), highlighting its significant growth potential. However, beyond hardware and regulatory challenges, the major obstacle lies in managing and scaling delivery operations. Therefore, businesses require a specialized management platform capable of automating workflows and optimizing the entire drone delivery lifecycle.

1.3 Project Introduction

Motivated by the challenges mentioned above, the team aims to develop SmartDroneDelivery—an AI-powered drone delivery management platform. The system focuses on managing the entire delivery lifecycle through five core business processes, including delivery request management, package processing and preparation, delivery operation management, real-time shipment tracking, and delivery confirmation.

The key feature of the project is the integration of Artificial Intelligence (AI) as an intelligent assistant to optimize operational efficiency. The system provides functionalities such as Estimated Time of Arrival (ETA) prediction, order status summarization, AI-powered chatbot support, and intelligent reporting and analytics.

1.4 Project Objectives

- **Platform Development**: Develop an AI-powered drone delivery management platform to improve operational efficiency and optimize logistics processes.
- **Business & Operations:** Automate the five core processes of the delivery lifecycle, from delivery request creation to delivery confirmation.
- **AI Integration:** Leverage AI for ETA prediction, order status summarization, chatbot-assisted customer support, and analytical reporting.
- **Technical Architecture**: Build a standardized, scalable software architecture capable of real-time data processing.
- **User Experience:** Provide an intuitive and user-friendly interface for both administrators and end users.

1.5 Project Scope

Functional Scope:

- In-Scope: Research, analyze, design, and develop the software platform for drone delivery management operations.
- Out-of-Scope: Drone hardware design and assembly, and flight control system development.

Data & Experimentation: Evaluate system feasibility and machine learning model accuracy using simulated datasets in small to medium-scale scenarios.

2 Project Management Plan

2.1 Development Methodology

- **Start Date: 31 July 2026**
- **Expected Completion Date:**
- **Project Management Methodology:** The project adopts the Scrum framework to manage and develop the software. Scrum divides the project into multiple short sprints, allowing the team to effectively monitor progress, respond to changing requirements, and evaluate the results after each sprint.
- **Jira Workflow:** The team uses Jira to manage project tasks throughout the development process.

- The workflow consists of the following stages:

To Do → In Progress → Done

Each task is created as a Jira Issue and assigned to the responsible team member. The issue status is updated according to the progress of the task.

- **Technology Stack**

Category	Technology / Tools
IDEs / Editors	Visual Studio Code, Visual Studio
Diagramming	Draw.io, Visual Paradigm, Lucid app
Version Control	GitHub
Project Management	Jira
Documentation	Latex

2.2 Project Planning

#	Sprint	Due Date	Main Tasks
1	**Sprint 1**	31/07 – 03/08	• Project analysis • SRS documentation • Actor identification, Use Case analysis • Business Process Analysis (BPMN) • Overview Architecture
2	**Sprint 2**	04/08 – 09/08	• Activity Diagram • Sequence Diagram • ERD design • DataBase Design

2.3 Configuration Management

The project uses GitHub for source code management and Jira for task management.

Working Process

Step 1. Clone the Repository

Each team member clones the GitHub repository to their local machine.

git clone `https://github.com/YaFhun06/CNPM.git`

Step 2. Update the Jira Status

Select the assigned task and change its status from:

$$\textit{To Do} \rightarrow \textit{In Progress}$$

Step 3. Create a Working Branch

Team members must not work directly on the main branch.

Create a feature branch following the team's naming convention.

Branch Naming Convention: *feature/sprint1-member-name*

Example:

feature/sprint1-phung

feature/sprint2-bao

Step 4. Implement the Assigned Tasks

Each member works on the assigned tasks within the project structure using Visual Studio Code.

Step 5. Commit Changes

Commit messages should reference the corresponding Jira Issue Key.

Commit Convention: *<Jira Issue Key>: <Action> <Description>*

Where:

- Jira Issue Key: CNPM-3, CNPM-15, …
- Action: Add, Update, Fix, Refactor, Remove, Complete
- Description: Brief description of the changes

Example: *git commit -m "CNPM-3: Add use case list"*

Step 6. Push the Working Branch

git push origin feature/sprint1-phung

Step 7. Update the Jira Status

After the task has been completed and confirmed by the team, change its status from:

$$\textit{In Progress} \rightarrow \textit{Done}$$

3 Software Requirement Specification

3.1 Functional Requirements

Module	Functions
Module 1. User Management	1. Register an account. 2. Log in and log out. 3. Reset forgotten passwords. 4. Manage user profiles 5. Manage user roles. 6. Manage user permissions.
Module 2. Customer & Delivery Management	1. Manage customer information. 2. Manage delivery addresses. 3. Create, view, update, and cancel delivery orders. 4. Track delivery order status. 5. Manage package information (weight, dimensions, description, etc.).

Module	Functions
Module 3. Delivery Workflow Management	1. Review and approve delivery orders. 2. Reject delivery orders. 3. Schedule deliveries. 4. Assign orders to landing stations. 5. Monitor delivery progress. 6. Update delivery status. 7. Handle failed deliveries. 8. Complete delivery orders.
Module 4. Landing Station Management	1. Manage landing stations. 2. Manage landing station locations and operational status. 3. Monitor landing station capacity. 4. Confirm package receipt. 5. Confirm drone pickup and return. 6. Update package status at the landing station.

Module	Functions
Module 5. Dashboard & AI Services	1. Display an overview dashboard. 2. View delivery order statistics. 3. View revenue statistics. 4. Monitor successful and failed delivery rates. 5. Monitor landing station performance. 6. Predict Estimated Time of Arrival (ETA). 7. Provide AI chatbot support. 8. Generate AI delivery summaries. 9. Perform AI-powered analytics and delivery trend analysis.

3.2 Non-Functional Requirements

- **NFR-01 Security & Access Control:** The system implements Role-Based Access Control (RBAC), secure authentication via JWT/OAuth 2.0, and end-to-end data encryption for sensitive information both at rest (AES-256) and in transit (HTTPS/TLS).
- **NFR-02 Performance & Scalability:** The system delivers high performance with API response times under 2 seconds for 95% of standard requests, maintaining high throughput and concurrency during peak user traffic.
- **NFR-03 Interoperability & Integration:** The platform supports seamless integration with third-party systems (e.g., LMS, OAuth providers, cloud storage) using secure, standardized RESTful APIs.
- **NFR-04 Containerization & Deployment:** All system components are fully containerized using Docker, ensuring consistent deployment environments across Development, Staging, and Production.
- **NFR-05 Cross-Platform & Usability**: The application features a fully responsive UI design, ensuring optimized user experience and seamless operation across desktop browsers and mobile devices.

- **NFR-06 System Monitoring:** The platform integrates real-time health checks and performance monitoring tools to continuously track server status, CPU/RAM utilization, and system anomalies.
- **NFR-07 Audit Logging & Data Backup:** Comprehensive audit logging records critical user activities for traceability, while automated database backup procedures run periodically to ensure data integrity and disaster recovery.
- **NFR-08 Maintainability & Modularity:** The codebase is structured following clean, modular software architecture standards, allowing for straightforward maintenance, debugging, and future updates.

3.3 User Requirement

3.3.1 Actor

Actor	Description
Customer	Creates delivery requests, tracks orders in real time, confirms delivery, and interacts with the AI Chatbot for inquiry and ETA prediction.
Dispatcher	Schedules flights, assigns orders to drones, monitors overall dispatching, and resolves delivery issues using AI suggestions.
Station Operator	Inspects and confirms package handovers at stations, updates order status, and manages station capacity/operational status.
Logistics Manager	Manages user accounts, configures Role-Based Access Control (RBAC), handles audit logs, backups, and ensures platform security and stability.
System Administrator	Oversees system performance via Dashboard analytics and uses AI-generated reports to optimize logistics efficiency.

3.3.2 Use Case Diagram

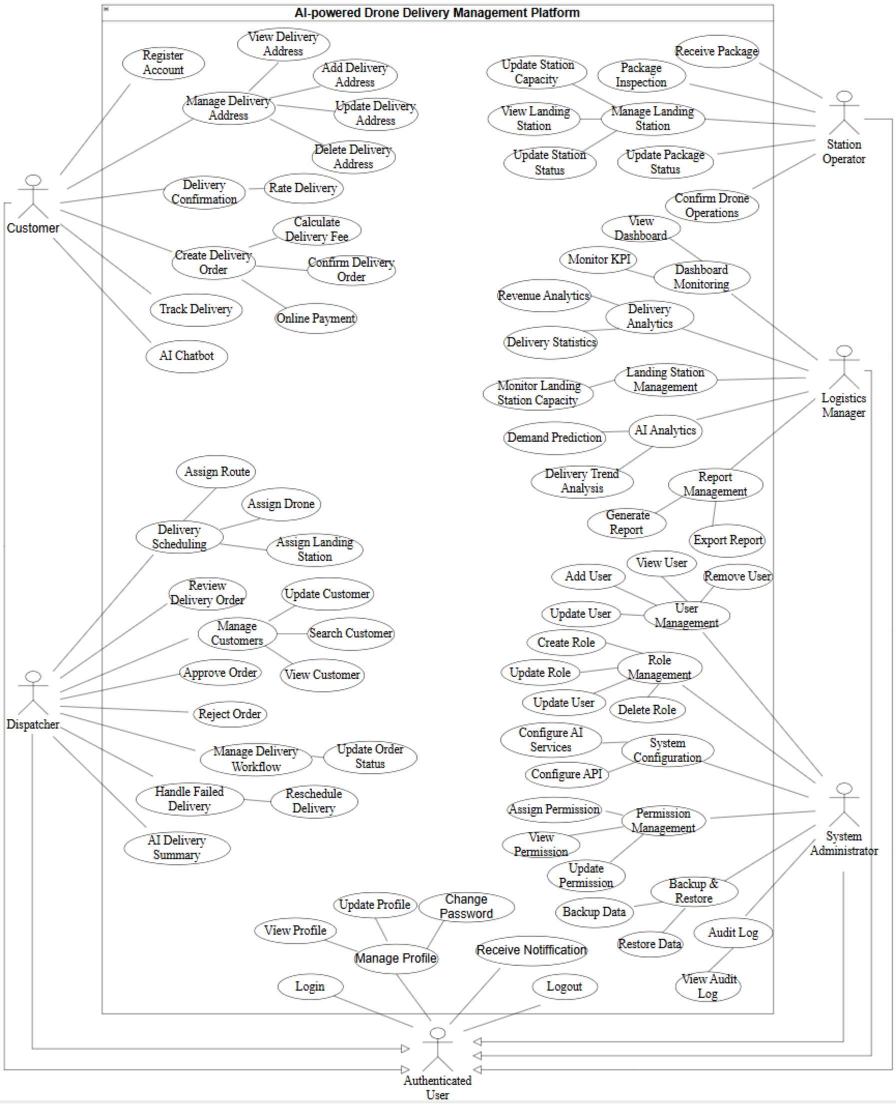

Figure 1: Use Case Diagram

3.3.3 Use Case Descriptions

ID	Use Case	Actor	Use Case Description
UC01	Register Account	Customer	Create a new customer account using an email address or phone number.
UC02	Login	Authenticated User	Authenticate and log into the system using registered credentials.
UC03	Logout	Authenticated User	Log out of the system securely.

ID	Use Case	Actor	Use Case Description
UC04	Forgot Password	Customer	Request a password reset through email or phone verification.
UC05	View Profile	Authenticated User	View personal account information.
UC06	Update Profile	Authenticated User	Update personal profile information such as name, phone number, or avatar.
UC07	Change Password	Authenticated User	Change the account password securely.
UC08	Receive Notification	Authenticated User	Receive notifications about delivery status, system alerts, and important updates.
UC09	Manage Delivery Address	Customer	Manage saved delivery addresses.
UC10	View Delivery Address	Customer	View all saved delivery addresses.
UC11	Add Delivery Address	Customer	Add a new delivery address.
UC12	Update Delivery Address	Customer	Edit an existing delivery address.
UC13	Delete Delivery Address	Customer	Remove a saved delivery address.
UC14	Create Delivery Order	Customer	Create a new drone delivery request.
UC15	Calculate Delivery Fee	Customer	Calculate the estimated delivery fee based on distance, package weight, and service type.
UC16	Confirm Delivery Order	Customer	Review and confirm the delivery request before submission.
UC17	Online Payment	Customer	Pay for the delivery order using supported payment methods.
UC18	Manage Own Orders	Customer	Manage personal delivery orders before they are completed.
UC19	View Orders	Customer	View current and historical delivery orders.
UC20	Edit Order	Customer	Modify a delivery order before it is approved by the dispatcher.

ID	Use Case	Actor	Use Case Description
UC21	Cancel Order	Customer	Cancel a delivery order before processing begins.
UC22	Track Delivery	Customer	Track the real-time delivery status, drone location, and estimated arrival time (ETA).
UC23	Delivery Confirmation	Customer	Confirm that the package has been received successfully.
UC24	Rate Delivery	Customer	Rate the delivery service after receiving the package.
UC25	AI Chatbot	Customer	Interact with the AI chatbot for delivery support, FAQs, and inquiries.
UC26	Manage Customers	Dispatcher	Manage customer information within the delivery system.
UC27	View Customer	Dispatcher	View customer information and delivery history.
UC28	Search Customer	Dispatcher	Search for customers by name, phone number, or account ID.
UC29	Update Customer	Dispatcher	Update customer information when necessary.
UC30	Review Delivery Order	Dispatcher	Review newly submitted delivery orders before processing.
UC31	Approve Order	Dispatcher	Approve a delivery request for scheduling and execution.
UC32	Reject Order	Dispatcher	Reject a delivery request that does not meet requirements.
UC33	Delivery Scheduling	Dispatcher	Schedule approved delivery orders.
UC34	Assign Drone	Dispatcher	Assign an available drone to a delivery order.
UC35	Assign Landing Station	Dispatcher	Select the appropriate landing station for package handling.
UC36	Assign Route	Dispatcher	Assign the optimal delivery route for the drone.
UC37	Manage Delivery Workflow	Dispatcher	Monitor and manage the delivery workflow throughout the delivery lifecycle.

ID	Use Case	Actor	Use Case Description
UC38	Update Order Status	Dispatcher	Update the delivery order status during processing.
UC39	Handle Failed Delivery	Dispatcher	Handle delivery failures by rescheduling or returning the package.
UC40	Reschedule Delivery	Dispatcher	Create a new delivery schedule for failed deliveries.
UC41	AI Delivery Summary	Dispatcher	View AI-generated delivery summaries and recommendations.
UC42	View ETA Prediction	Dispatcher	View AI-predicted estimated delivery time for active orders.
UC43	Receive Package	Station Operator	Receive packages from customers or logistics staff at the landing station.
UC44	Package Inspection	Station Operator	Inspect the package for weight, size, condition, and labeling before dispatch.
UC45	Manage Landing Station	Station Operator	Manage the operational status and capacity of the landing station.
UC46	View Landing Station	Station Operator	View landing station information and current operational status.
UC47	Update Station Status	Station Operator	Update the landing station status (Available, Busy, Maintenance, etc.).
UC48	Update Station Capacity	Station Operator	Update the available capacity of the landing station.
UC49	Update Package Status	Station Operator	Update the package status during handling and delivery.
UC50	Confirm Drone Operations	Station Operator	Confirm package loading and drone readiness before takeoff.
UC51	Dashboard Monitoring	Logistics Manager	Monitor overall delivery operations through dashboards.
UC52	View Dashboard	Logistics Manager	View key operational metrics and system status.
UC53	Monitor KPI	Logistics Manager	Monitor key performance indicators (KPIs) such as delivery success rate and average delivery time.
UC54	Delivery Analytics	Logistics Manager	Analyze delivery performance and operational statistics.

ID	Use Case	Actor	Use Case Description
UC55	Revenue Analytics	Logistics Manager	Analyze delivery revenue and business performance.
UC56	Delivery Statistics	Logistics Manager	View statistical reports on deliveries and operational activities.
UC57	AI Analytics	Logistics Manager	View AI-generated insights and predictive analytics.
UC58	Demand Prediction	Logistics Manager	View AI predictions for future delivery demand.
UC59	Delivery Trend Analysis	Logistics Manager	Analyze delivery trends over time.
UC60	Landing Station Management	Logistics Manager	Monitor and manage landing stations across the system.
UC61	Monitor Landing Station Capacity	Logistics Manager	Monitor the operational capacity of all landing stations.
UC62	Report Management	Logistics Manager	Manage operational reports for the logistics system.
UC63	Generate Report	Logistics Manager	Generate operational, analytical, and performance reports.
UC64	Export Report	Logistics Manager	Export reports in PDF, Excel, or CSV format.
UC65	User Management	System Administrator	Manage user accounts within the system.
UC66	View User	System Administrator	View user account information.
UC67	Add User	System Administrator	Create a new user account.
UC68	Update User	System Administrator	Update existing user information.
UC69	Remove User	System Administrator	Remove or deactivate a user account.
UC70	Role Management	System Administrator	Manage system roles.
UC71	Create Role	System Administrator	Create a new role.
UC72	Update Role	System Administrator	Modify an existing role.

ID	Use Case	Actor	Use Case Description
UC73	Delete Role	System Administrator	Delete an existing role.
UC74	Assign Role	System Administrator	Assign roles to users.
UC75	Permission Management	System Administrator	Manage permissions for system roles.
UC76	View Permission	System Administrator	View permission settings.
UC77	Assign Permission	System Administrator	Assign permissions to roles.
UC78	Update Permission	System Administrator	Modify permission settings.
UC79	System Configuration	System Administrator	Configure system settings, APIs, notifications, and AI services.
UC80	Audit Log	System Administrator	View and monitor system logs and user activities.
UC81	View Audit Log	System Administrator	View detailed audit log records.
UC82	Backup & Restore	System Administrator	Manage system backup and data restoration.
UC83	Backup Data	System Administrator	Perform a system data backup.
UC84	Restore Data	System Administrator	Restore system data from a backup file.

4 Software Design Description

4.1 Business Process

4.1.1 Business Process Description

The delivery process of the system consists of the following steps:

Step 1. Create a Delivery Order: The customer logs into the system, enters the sender's and recipient's information, pickup and delivery addresses, package details, and submits a delivery request.

Step 2. Review and Approve the Delivery Order: The system validates the submitted information. The Dispatcher reviews the delivery request and decides whether to approve or reject it. If the request is rejected, the system notifies the customer and provides the reason for rejection.

Step 3. Schedule the Delivery and Assign a Drone: The Dispatcher selects an appropriate Landing Station and schedules the delivery. The Drone Delivery System then assigns a suitable drone based on its location, payload capacity, battery level, and operational status.

Step 4. Receive the Package: The Station Operator receives and inspects the package. If the package meets the transportation requirements, it is handed over to the drone, and the system updates the delivery status to Package Received or Ready for Delivery.

Step 5. Deliver the Package: The drone delivers the package along the optimized route. During the delivery process, the system continuously updates the order status, drone location, and the Estimated Time of Arrival, allowing customers to track their delivery in real time.

Step 6. Complete the Delivery: Once the recipient confirms receipt of the package, the system updates the order status to Delivered, records the delivery information, and allows the customer to provide service feedback.

Step 7. Handle Delivery Failure: If the delivery cannot be completed due to technical issues, adverse weather conditions, or other unexpected situations, the system updates the order status to Delivery Failed, notifies the customer, and the Dispatcher determines the appropriate resolution, such as rescheduling the delivery, returning the package, or canceling the order.

4.1.2 BPMN Diagram

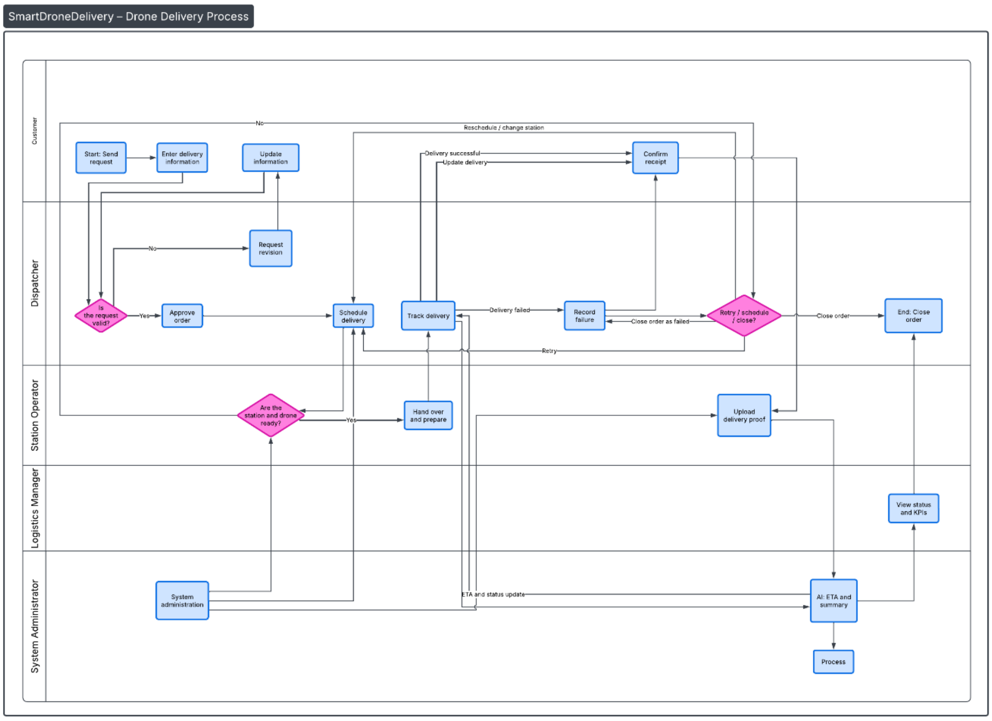

Figure 2: BPMN Diagram of the Delivery Process

4.2 Activity Diagram

4.3 Sequence Diagram

4.4 Software Architecture

4.4.1 Architecture Overview

SmartDroneDelivery adopts a Modular Monolith architecture combined with Clean Architecture for the backend and an independent AI service layer. This architecture provides a clear separation of concerns, simplifies maintenance, supports future scalability, and enables efficient deployment through Docker.

4.4.2 Architecture Diagram

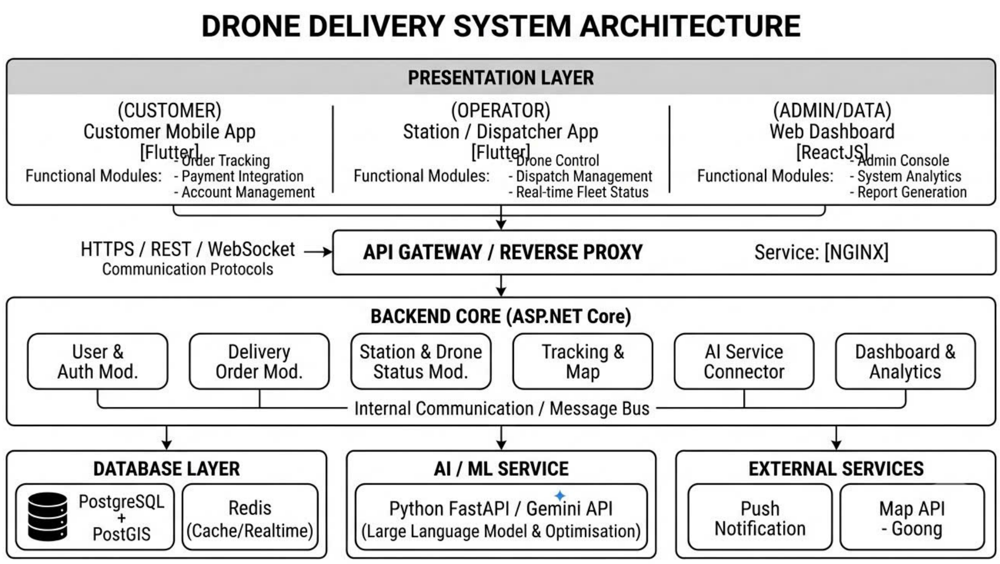

Figure 3: Software Architecture Diagram

4.4.3 Module Design

4.4.4 Project Structure

4.5 UI/UX Design

4.5.1 User Flow

4.5.2 Navigation Structure

4.5.3 Wireframe

4.5.4 Prototype

4.5.5 Prototype

4.6 REST API Design

References

1. "Định nghĩa: Nền tảng giao hàng bằng drone là gì (Drone Delivery Digital Platforms)," [Online]. Available: https://fmit.vn/tu-dien-quan-ly/drone-delivery-digital-platforms-la-gi. [Accessed: Aug. 2, 2026].

2. "Xu hướng phát triển và tiềm năng của drone trong ngành Logistics – Ứng dụng Drone trong Logistics: Vận chuyển thông minh & hiệu quả," [Online]. Available: https://sundrone.vn/ung-dung-drone-trong-logistic/. [Accessed: Aug. 2, 2026].

3. "Thị trường hiện tại và triển vọng trong tương lai của máy bay không người lái (Drone) – Dịch vụ giao hàng bằng drone: Cách mạng hóa ngành logistics," [Online]. Available: https://vnfocus.vn/2025/06/20/dich-vu-giao-hang-bang-drone-cach-mang-hoa-nganh-logistics/. [Accessed: Aug. 3, 2026].

4. "Artificial Intelligence in Logistics and Distribution: The Function of AI in Dynamic Route Planning for Transportation, Including Self-Driving Trucks and Drone Delivery Systems," [Online]. Available: https://www.researchgate.net/profile/Godwin-Aipoh/publication/389430364_Artificial_Intelligence_in_Logistics_and_Distribution_The_function_of_AI_in_dynamic_route_planning_for_transportation_including_self-driving_trucks_and_drone_delivery_systems/links/67e2151d35f7044c92865ed1/Artificial-Intelligence-in-Logistics-and-Distribution-The-function-of-AI-in-dynamic-route-planning-for-transportation-including-self-driving-trucks-and-drone-delivery-systems.pdf. [Accessed: Aug. 2, 2026].